



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 03

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Objective of Lab No. 03:

After performing lab 3, students will be able to:

- o Understand and apply lists and nested list
- o Understand and apply tuples
- o Understand and apply dictionaries
- o Use Above concepts to solve problems

Lab Tasks:

Task#01: Student Attendance & Performance Analyzer (With Grades)

- a. Take Number of students as input from the user.
- b. For each student take the following input:
 - i. Name
 - ii. Attendance
 - iii. Marks
 - iv. Check the Grade of student based on marks, the grades are either ("A","B","C","F")
 - v. Now store the data in dictionary, where key is the student name, and value is another dictionary, which contains attendance, marks and grade of student as a key value pair.
 - vi. Append the marks for each student to a list



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- c. Now Iterate over the data dictionary and display names of those students who are eligible for exams.
- d. Iterate over the marks list and find out average marks and fail count (if marks are below 50)

Sample outputs are shown below:

```
Enter number of students: 3

Enter name: Ali
Enter attendance %: 82
Enter marks: 88

Enter name: Sara
Enter attendance %: 70
Enter marks: 65

Enter name: Ahmed
Enter attendance %: 90
Enter marks: 42
```

```
Eligible students for exam:

Ali - A
Ahmed - Fail

Class Average Marks: 65.0
Number of Failed Students: 1
```

Lab Task 2: Daily Expense Analyzer

Scenario

A user wants to analyze daily expenses for better budget

- a. Ask user to enter expenses until they type "done"
- b. Store expenses in a **list**
- c. Use **loops** to calculate:
 - o total expense



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- average expense
- d. Use **list comprehension** to find expenses above average
- e. Display:
 - highest and lowest expense
- f. Use a **tuple** to store (total, average)

Lab Task 3:

You are given a list of **student feedback comments** about a course.

Each comment is a **string** containing words like *good*, *excellent*, *average*, *bad*, or *poor*.

Your task is to analyze this feedback using **lists**, **loops**, **conditionals**, and **string functions**.

feedbacks = [

"The course was excellent and very good",
"Good content but the pace was average",
"The lectures were poor and bad",
"Excellent explanation and good examples",
"Average course but good instructor"

]

Normalize the Feedback

- Convert each feedback to **lowercase**
- Remove **extra spaces** (if any)

Keyword Counting

Using a loop and conditionals, count how many times each of the following words appears:

- excellent
- good



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- average
- bad
- poor

Store the counts in **dictionary**.

Feedback Classification

For each feedback:

- If it contains "**excellent**" or "**good**", classify it as **Positive**
- If it contains "**average**", classify it as **Neutral**
- If it contains "**bad**" or "**poor**", classify it as **Negative**

Print the feedback along with its category.

Summary Report:

After processing all feedbacks, print:

- Total number of **Positive**, **Neutral**, and **Negative** feedbacks
- The **most frequently used word** among the keywords



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```
Feedback: The course was excellent and very good
Category: Positive
```

```
Feedback: Good content but the pace was average
Category: Neutral
```

```
...
```

```
Summary:
```

```
Positive Feedbacks: 3
```

```
Neutral Feedbacks: 1
```

```
Negative Feedbacks: 1
```

```
Most Frequent Word: good
```

Lab Task#04:

Ride-Hailing Fare Calculator

1. Store Ride Types in a Dictionary: "mini": 25, "sedan": 40, "suv": 60
2. Store peak hours in a tuple, peak hours are: (18, 19, 20, 21)
3. Make an empty list named rides to store all ride records
4. Run a loop:
 - a. Accept Distance in km, if the value is zero break the loop
 - b. Enter the hour (0-23)
 - c. Enter the ride type (mini, sedan, suv)
 - d. Retrieve the rate from the dictionary based on ride type
 - e. Calculate the fare
 - f. If the hour was in peak hours, apply surge of 20%



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- g. Store ride details: distance, time, ride_type, fare in a dictionary
 - h. Append this dictionary to rides list.
5. Calculate Total fare which organization has earned.